

Courses in Econ Department

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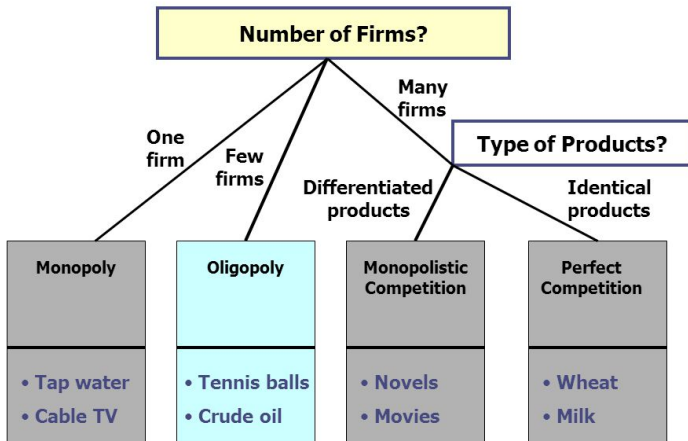
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What do we learn in microeconomics?

- ▶ Consumers' behaviors
- ▶ Firm theory
- ▶ Competitive market equilibrium
- ▶ Monopoly
- ▶ Oligopoly and monopolistic competition
- ▶ Game theory
- ▶ Welfare economics and general equilibrium
- ▶ Public goods and externality

Different market structure

The Four Types of Market Structure



Game theory example

Possible Outcomes of Prisoner's Dilemma

Outcome	Henry Cooperates	Henry Defects
Elizabeth Cooperates	(1,1)	(5,0)
Elizabeth Defects	(0,5)	(3,3)

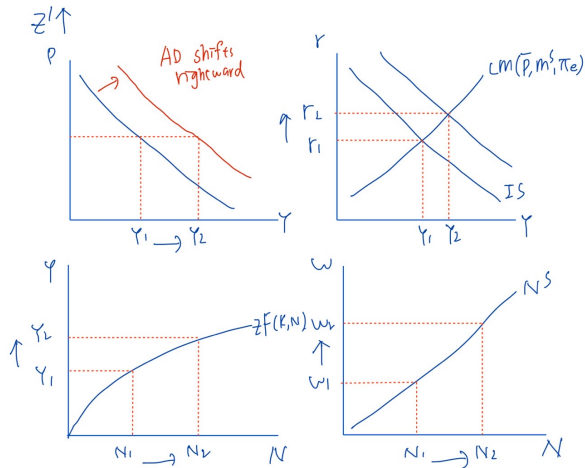
Penalties for (Elizabeth, Henry)

In this case, each robber always has an incentive to defect, regardless of the choice the other makes.

What do we learn in macroeconomics?

- ▶ Solow growth model
- ▶ Intertemporal consumption decision
- ▶ Equilibrium in an endowment economy
- ▶ Neoclassical business cycle model
- ▶ New Keynesian model

Analysis of change

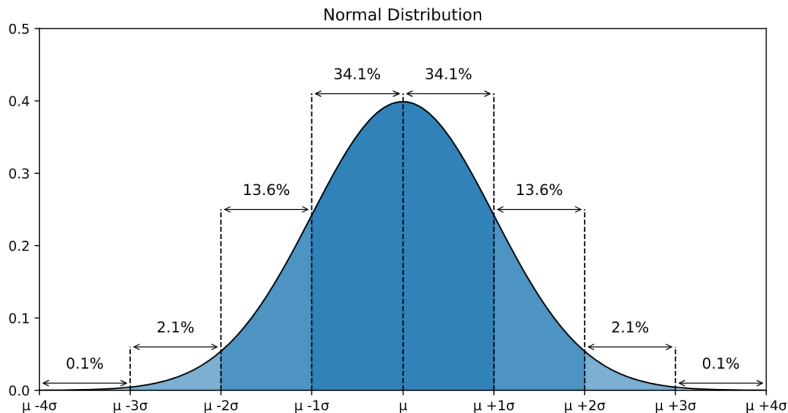


IS, AD shifts rightward
 $Y \uparrow$ $r \uparrow$ $N \uparrow$ $w \uparrow$

What do we learn in statistics?

- ▶ Descriptive statistics
- ▶ Random variables
- ▶ Asymptotic theory
- ▶ Estimation
- ▶ Hypothesis testing

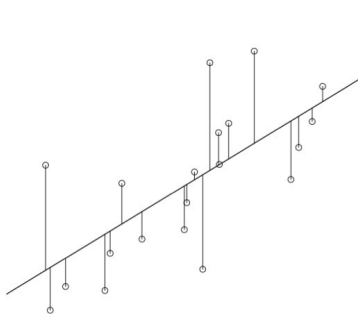
Normal distribution



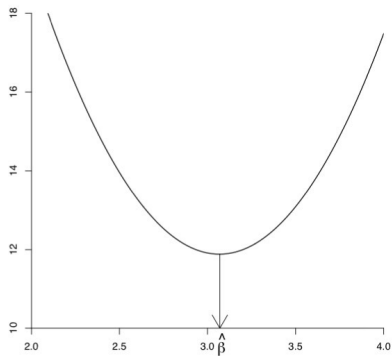
What do we learn in econometrics?

- ▶ Roy model
- ▶ Linear regression
- ▶ Instrumental variables (IV)
- ▶ Difference-in-Differences (DiD)
- ▶ Regression discontinuity design (RDD)

Method of least squares



(a) Deviation from Fitted Line



(b) Sum of Squared Error Function

Figure 3.1: Regression With One Regressor

References

- ▶ <https://symphysismarketing.com/2019/04/27/the-four-types-of-market-structure/>
- ▶ <https://www.investopedia.com/terms/p/prisoners-dilemma.asp>
- ▶ https://www.w3schools.com/statistics/statistics_normal_distribution.php
- ▶ Hansen, B. (2022). Econometrics. Princeton University Press.